

# Vacation-Rental Marketplace — Production Architecture

Scaling strategy for frontend, backend, storage, search & deployment · Airbnb-style listing platform

■ Edge / Delivery   ■ Frontend   ■ Backend services   ■ Storage & search   ■ Async & data platform   ■ Cross-cutting

## CLIENTS

### Web (Next.js)

Desktop & mobile web, this app

### iOS / Android

React Native / native

### Host dashboard

Listing & calendar mgmt

### Partners / API

OTA channel integrations

## EDGE & DELIVERY

### Global CDN

Static assets, image CDN, HLS

### Edge network / WAF

TLS, DDoS, bot mgmt

### Edge SSR/ISR

Vercel / CloudFront functions

### API Gateway

Routing, authN, rate limit

### GraphQL / BFF

Client-shaped aggregation

## FRONTEND PLATFORM

### Next.js App Router

SSR listing pages + ISR for SEO, streamed RSC

### Design system

Shared component lib, tokens, a11y primitives

### Client state / cache

React Query, optimistic booking UI

## BACKEND MICROSERVICES

### Listings service

CRUD, photos, amenities

### Search service

Geo + filters, ranking

### Availability & Pricing

Calendar, dynamic pricing

### Booking / Reservations

Holds, idempotent txns

### Payments & Payouts

Stripe, escrow, FX

### Reviews & Ratings

Guest favourite scoring

### Users & Auth

Identity, OAuth, KYC

### Messaging & Notifications

Chat, email, push

## STORAGE & SEARCH

### PostgreSQL

Sharded by region; listings, bookings

### Redis

Sessions, hot cache, locks

### Elasticsearch

Full-text + geo search

### Object store (S3)

Photos, originals

### Time-series

Pricing / metrics

### Data warehouse

Snowflake / BigQuery